HKUST – Department of Computer Science and Engineering
COMP 2711: Discrete Math Tools for Computer Science
Spring 2025 Final Examination

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This is a closed-book exam. You are allowed to bring two A4-sized cheat sheets, and use a calculator.
- Each question is on a separate page. This is for clarity and is not meant to imply that each question requires a full-page answer. Some can be answered using only a few lines. You may also use the back of the pages if needed.
- Solutions can be written in terms of binomial coefficients, factorials, and the $C(n,k)$, $P(n,k)$ notation. For example, you can write $\binom{5}{3} + \binom{4}{2}$ instead of 16. Avoid using nonstandard notations such as ${}_nP_k$ or ${}_nC_k$.
- Please let us know if you need more scratch papers.
- Please sign the following honor code.

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I declare that the answers submitted for
this examination are my own work.

I understand that sanctions will be
imposed, if I am found to have violated the
University regulations governing academic
integrity.

Student's Name: _____

Student's Signature: _____
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Problems	1	2	3	4	5	6	7	8	9	10	11	12	Total
Points	9	6	8	6	15	10	6	7	7	7	7	12	100
Score													

Problem 1: Logic, Set and Function [9 pts]

(a) **[3 pts, 1 each]** Determine whether the following statements are True or False. No argument is needed. $\mathcal{P}(X)$ denotes the power set of X.

(i) There exists a countably infinite set X with $\mathcal{P}(X)$ uncountable.

(ii) There exists a countably infinite set X with $\mathcal{P}(X)$ also countably infinite.

(iii) There is a bijection between all the real numbers of $[0, 1]$ and all the real numbers of $[0, 1)$.

(b) **[2 pts]** Prove or disprove the following statement. In order to prove, you must use propositional laws. In order to disprove, you need to come up with values that this statement is false.

$$(x \vee y \vee \neg z) \wedge (\neg x \vee y \vee z) \equiv \neg y \rightarrow (x \leftrightarrow z)$$

(c) **[2 pts]** Assume A and B are two sets with elements in $\mathbb{N}$. Express the following predicate using only $\forall$, $\exists$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, $\neg$, $\in$, $\notin$, parentheses, variables and constants. Take the natural numbers $\mathbb{N}$ as the domain. You are not allowed to use any other predicates.

Set A and set B are disjoint

(d) **[2 pts]** Express the negation of the statement in (c) so that all negation symbols, if any, must immediately precede predicates.

Answer: (a) i)T ii)F iii)T

(b) $(x \vee y \vee \neg z) \wedge (\neg x \vee y \vee z) \equiv y \vee ((x \vee \neg z) \wedge (\neg x \vee z)) \equiv y \vee ((z \rightarrow x) \wedge (x \rightarrow z)) \equiv y \vee (x \leftrightarrow z) \equiv \neg y \rightarrow (x \leftrightarrow z)$

(c) Possible Solutions:

* $\forall x(x \in A \rightarrow x \notin B) \wedge \forall x(x \in B \rightarrow x \notin A)$
* $\forall x(x \notin A \vee x \notin B)$
* $\neg\exists x(x \in A \wedge x \in B)$

(d) $\exists x(x \in A \wedge x \in B)$

Problem 2: Number Theory, Fermat's Little Theorem [6 pts]

(a) **[3 pts]** Calculate $2^{62} \pmod{63}$.

(b) **[3 pts]** Use **Fermat's Little Theorem** to prove that 63 is not a prime number. Note that simply factorizing 63, e.g., as 3×21, is not a valid solution here. You must apply Fermat's Little Theorem explicitly.

Hint: You might find part (a) useful here.

Answer: (a) $2^6 \equiv 1 \pmod{63} \implies 2^{60} \equiv 1 \implies 2^{62} \equiv 4 \pmod{63}$

(b) We know that if p is prime, Then $a^{p-1} \equiv 1 \pmod{p}$. (if a is not a multiple of p)

So we can conclude that $a^{p-1} \not\equiv 1 \pmod{p} \rightarrow p$ is not prime.

As $2^{62} \not\equiv 1 \pmod{63}$ we can conclude that 63 is not prime!

Problem 3: Chinese Remainder Theorem, Counting [8 pts]

For a positive integer p, define the positive integer n to be p-*safe* if n differs in absolute value by more than 2 from all multiples of p. For example, the set of 10-*safe* numbers are $3, 4, 5, 6, 7, 13, 14, 15, 16, 17, \cdots$.

(a) **[2 pts]** Let x be a 7-*safe* number. What are the possible values of x mod 7?

(b) **[6 pts]** Find the number of positive integers less than 1001 ($1001 = 7 \cdot 11 \cdot 13$) that are simultaneously 7-*safe*, 11-*safe*, and 13-*safe*. Please justify your answer.

Answer: (a) 3,4

(b) Consider the mapping from x to the 3-tuple ($a = x \bmod 7, b = x \bmod 11, c = x \bmod 13$). For any x that is 7-safe, 11-safe, and 13-safe, we must have $a \in \{3, 4\}, b \in \{3, 4, 5, 6, 7, 8\}, c \in \{3, 4, 5, 6, 7, 8, 9, 10\}$. By the Chinese Remainder Theorem, for each such triple (a, b, c), there is also a unique positive integer $x < 1001$ such that $a = x \bmod 7, b = x \bmod 11, c = x \bmod 13$. So there is a one-to-one correspondence between such triples and the set of positive integers less 1001 that are 7-safe, 11-safe, and 13-safe. Thus, the answer is

$$2 \cdot 6 \cdot 8 = 96.$$

Problem 4: Number Theory, Counting [6 pts]

The least common multiple (LCM) of two integers x and y is the smallest positive integer that is a multiple of both x and y. As an example, the least common multiple of 8 and 10 is 40.

(a) **[1 pt]** Assume $x = 2^3 \cdot 5^6$ and $y = 2^4 \cdot 5^3$. What is the least common multiple of x and y?

(b) **[5 pts]** Determine the number of ordered pairs of positive integers (a, b) such that the least common multiple of a and b is $2^3 \cdot 5^7 \cdot 11^{13}$. Briefly explain your answer.

Answer: (a) Their LCM is $2^{max(3,4)} \cdot 5^{max(6,3)} = 2^4 \cdot 5^6$

(b) Both a and b are divisors of $2^3 5^7 11^{13}$, and so a = $2^x 5^y 11^z$ and b = $2^s 5^t 11^u$ for some nonnegative integers x, y, z, s, t, u. Because $2^3 5^7 11^{13}$ is the least common multiple, $\max(x, s) = 3$, $\max(y, t) = 7$, and $\max(z, u) = 13$. Hence (x, s) can be equal to $(0, 3), (1, 3), (2, 3), (3, 3), (3, 2), (3, 1)$, or $(3, 0)$, so there are 7 choices for (x, s). Similarly, there are 15 and 27 choices for (y, t) and (z, u), respectively. By the product rule, there are $7 \times 15 \times 27 = 2835$ ordered pairs of positive integers (a, b) having $2^3 5^7 11^{13}$ as their least common multiple.

Problem 5: Algorithm Analysis [15 pts]

(a) **[6 pts, 2 each]** For each pair of expressions (A, B), indicate whether A is O, Ω, Θ, of B. Note that zero, one, or more of these relations may hold for a given pair. **List all applicable relations. No explanation is needed.**

Examples: (a) for $A = n, B = 2n$, all O, Ω and Θ are correct; (b) for $A = n, B = n^2$, only O is correct.

(i) $A = 3^{\log_2(n^2)}$, $B = n^4$

(ii) $A = n \log n$, $B = \log((n!)^2)$

(iii) $A = n^3 + n^2(\log n)^4$, $B = 2^{\sqrt{n}}$

(b) **[4 pts]** Consider an array A with n elements. We call a pair (i, j) an *inversion*, if $i < j$ and $A[i] > A[j]$. The following algorithm counts the number of inversions for array A. What is its time complexity? Write your answer in the tightest big-O notation with respect to n.

Algorithm 1 CountInversions(n)

```
inversions = 0
for i = 1 to i = n − 1 do
    for j = i + 1 to j = n do
        if A[j] < A[i] then
            inversions = inversions + 1
        end if
    end for
end for
Return inversions
```

(c) **[5 pts]** Derive the tightest possible asymptotic upper bound for the following recurrence $T(n)$, assuming n is a power of a power of 2. Do not use the master theorem. Show all your steps to receive full credit.

$$T(2) = 1,$$
$$T(n) = \sqrt{n}T(\sqrt{n}) + n \quad \text{for } n > 2.$$

Answer: (a) i) O ii) O, Ω, Θ iii) O

(b) We can see that the for $i = 1$, the second loops cost is $n - 1$, for $i = 2$ the second loop cost is $n - 2$ and $\cdots$. So we can conclude that the total cost is $(n-1) + (n-2) + \cdots + 0 = \frac{n \cdot (n-1)}{2}$ which is $O(n^2)$

(c) By expansion:

$$
\begin{aligned}
T(n) &= n^{\frac{1}{2}} T(n^{\frac{1}{2}}) + n \\
&= n^{\frac{1}{2}} (n^{\frac{1}{4}} T(n^{\frac{1}{4}}) + n^{\frac{1}{2}}) + n \\
&= n^{\frac{1}{2}+\frac{1}{4}} T(n^{\frac{1}{2^2}}) + 2n \\
&= n^{\frac{1}{2}+\frac{1}{4}+\ldots+\frac{1}{2^k}} T(n^{\frac{1}{2^k}}) + kn \\
&= n^{\frac{1}{2}+\frac{1}{4}+\ldots+\frac{1}{2^k}} (1) + kn \qquad [\text{Set } n^{\frac{1}{2^k}} = 2, \text{we have } k = \log\log n] \\
&= n^{\frac{1/2}{1-1/2}} + kn \\
&= n + n \log\log n \\
&= O(n \cdot \log\log n)
\end{aligned}
$$

Solution: $O(n \cdot \log\log n)$

You can continue your solution for Problem 5 on this page.

Problem 6: Permutations and Combinations [10 pts]

(a) **[4 pts]** Nine chairs in a row are to be occupied by six students and Professors Alpha, Beta, and Gamma. These three professors arrive before the six students and decide to choose their chairs so that each professor will be between two students. In how many ways can Professors Alpha, Beta, and Gamma choose their chairs? Briefly explain why.

Answer: Imagine the six students standing in a row before they are seated. There are 5 spaces between them, each of which may be occupied by at most one of the 3 professors. Therefore, there are $P(5,3) = 5 \times 4 \times 3 = 60$ ways the professors can select their places.

(b) **[6 pts]** How many 6-digit natural numbers (from 100000 to 999999) have digits that sum to 11? For example, the number 100019 sums to $1+0+0+0+1+9 = 11$ and would be counted. Briefly explain why.

Answer: Let the 6-digit number be $\overline{a_1a_2a_3a_4a_5a_6}$, where $a_1 + a_2 + \cdots + a_6 = 11$ and $a_1 \geq 1$. Substitute $a_1 = a_1' + 1$, so $a_1' + a_2 + \cdots + a_6 = 10$, with $a_1', a_2, \ldots, a_6 \geq 0$. The number of non-negative solutions is $\binom{15}{5}$. Subtract invalid cases: 5 cases where any of $a_2, \ldots, a_6 = 10$, and 6 cases where $a_1' = 9$ or 10. Total valid numbers: $\binom{15}{5} - 11 = 2992$.

Problem 7: Pigeonhole Principle [6 pts]

A drawer in a dark room contains 100 red socks, 80 green socks, 60 blue socks and 40 black socks. A youngster selects socks one at a time from the drawer but is unable to see the color of the socks drawn. What is the smallest number of socks that must be selected to guarantee that the selection contains at least 10 pairs? (A pair of socks is two socks of the same color. No sock may be counted in more than one pair.) Suppose your answer is n. You need to argue that n socks guarantee at least 10 pairs, while $n-1$ socks are not enough.

Answer: **First Solution:** For any selection, at most one sock of each color will be left unpaired, and this happens if and only if an odd number of socks of that color is selected. Thus 24 socks suffice: at most 4 will be unpaired, leaving at least 20 in pairs. However, 23 will do! Since 23 is not the sum of four odd numbers, at most 3 socks out of 23 will be unpaired. On the other hand, 22 will not do: if the numbers of red, green, blue, and black socks are $5, 5, 5, 7$, then four are unpaired, leaving 9 pairs. Thus 23 is the minimum.

Second Solution: Proceed inductively. If we require only one pair, then it suffices to select 5 socks. Moreover, selecting 4 socks doesn't guarantee a pair since we might select one sock of each color. If we require two pairs, then it suffices to select 7 socks: any set of 7 socks must contain a pair; if we remove this pair, then the remaining 5 socks will contain a second pair as shown above. On the other hand, 6 socks might contain 3 greens, 1 black, 1 red and 1 blue - hence only one pair. Thus 7 socks is the smallest number to guarantee two pairs. Similar reasoning shows that we must draw 9 socks to guarantee 3 pairs, and in general, $2p+3$ socks to guarantee p pairs. This formula is easily proved by mathematical induction. Thus 23 socks are needed to guarantee 10 pairs.

Marking Scheme

- 4 pts for correct n
- 1 pt for argument that n is sufficient
- 1 pt for argument that $n-1$ is not sufficient
- If n is wrong, 1-2 pts.

Problem 8: Expected Value [7 pts]

Suppose that 5 boys and 9 girls are arranged randomly in a row. Let S be the number of adjacent pairs of a boy and a girl. For example, in the arrangement $BGGBGBBGGGBGGG$, we have $S = 7$. Find the expected value of S. Explain your solution.

Answer: **First Solution**: In general suppose there are k boys and $n-k$ girls. For $i = 1, 2, \ldots, n-1$, let A_i be the probability that there is a boy-girl pair in positions $(i, i+1)$ in the line. Since there is either 0 or 1 pair in $(i, i+1)$, A_i is also the expected number of pairs in these positions. By symmetry, all A_i's are the same (or note that the argument below is independent of i). Thus, the answer is $(n-1)A_i$.

We may consider the boys indistinguishable and likewise the girls. (Why?) Then an order is just a sequence of k Bs and $n-k$ Gs. To have a pair at $(i, i+1)$ we must have **BG** or **GB** in those positions, and the remaining $n-2$ positions must have $k-1$ boys and $n-k-1$ girls. Thus there are $2\binom{n-2}{k-1}$ sequences with a pair at $(i, i+1)$. Since there are $\binom{n}{k}$ sequences, the answer is

$$(n-1)A_i = (n-1)\frac{2\binom{n-2}{k-1}}{\binom{n}{k}} = \frac{2k(n-k)}{n}.$$

Thus, when $n = 14$ and $k = 5$, the answer is $2 \cdot 5 \cdot 9/14 = 90/14 = 45/7$.

Second Solution: There are 13 adjacent pairs in a row of 14 people. For any adjacent pairs, the probability that one is a boy and the other is a girl is the sum of two probabilities $P(BG) + P(GB)$.

$$P(BG) = \frac{5}{14} \cdot \frac{9}{13}; P(GB) = \frac{9}{14} \cdot \frac{5}{13}$$

$$P(BG \cup GB) = P(BG) + P(GB) = \frac{45}{91}$$

Since there are 13 adjacent pairs, and the probability of any pair being BG or GB is $\frac{45}{91}$. By the linearity of expectation for indicator random variables, the expected number of such pairs is: $13 \times \frac{45}{91} = \frac{45}{7}$. Also accept 6.428571429.

Marking Scheme

- 6 points – The solution is correct but contains a minor calculation error.
- 5 points – The solution is mostly correct but has a minor conceptual or procedural mistake.
- 3–4 points – The solution demonstrates a correct approach but is either incomplete or contains a major error.
- 1–2 points – The solution is incorrect but shows some understanding of the underlying concepts.

Problem 9: Counting, Hockey Stick Identity [7 pts]

Consider all 1000-element subsets of the set $\{1, 2, 3, \ldots, 2022\}$. From each such subset, choose the least element. Find the sum of all these least elements.

Hint: You can use the Hockey Stick Identity introduced in class without proving it:

$$\binom{n+1}{r+1} = \sum_{j=r}^{n} \binom{j}{r}$$

Answer:

- Number of subsets with 1 as the least element: $C(2021, 999)$
- Number of subsets with 2 as the least element: $C(2020, 999)$
- ...
- Number of subsets with 1023 as the least element: $C(999, 999)$

Sum of all least elements equals:

$$C(2021, 999) + 2 \cdot C(2020, 999) + \ldots + 1023 \cdot C(999, 999).$$

We can group the sum into columns, where each column can be computed using the Hockey Stick Identity:

$$\begin{aligned}
&C(2021, 999) + \\
&C(2020, 999) + C(2020, 999) + \\
&C(2019, 999) + C(2019, 999) + C(2019, 999) + \\
&\ldots \\
&C(999, 999) + C(999, 999) + C(999, 999) + C(999, 999) + \ldots
\end{aligned}$$

- 1st column sums to $C(2022, 1000)$
- 2nd column sums to $C(2021, 1000)$
- ...
- 1023rd column sums to $C(1000, 1000)$

We can again apply the Hockey Stick identity to the sum of all columns. This gives us $C(2023, 1001)$.

Problem 10: Inclusion-Exclusion Principle, Probability [7 pts]

A 3-by-3 unit-square grid is to be colored such that each unit square is independently colored red or blue with equal probability. What is the probability that the grid does not contain any 2-by-2 red square? Justify your answer.

Answer: Since each of the 9 unit squares of the 3-by-3 square can take one of two colors, in total there are 2^9 different ways of coloring the 3-by-3 square. Let us count now the colorings that form a 2-by-2 red square. This red square can be formed in the upper left, upper right, bottom left or bottom right corner of the 3-by-3 square. Let A, B, C, D be, respectively, the sets of the four types of colorings listed above that produce a 2-by-2 red square. Since the other 5 squares of the 3-by-3 square can have any color, we have $|A| = |B| = |C| = |D| = 2^5$. These sets are however *not* disjoint, so we must apply inclusion-exclusion:

$$\begin{aligned}
&|A \cup B \cup C \cup D| \\
&= |A| + |B| + |C| + |D| \\
&- |A \cap B| - |A \cap C| - |A \cap D| - |B \cap C| - |B \cap D| - |C \cap D| \\
&+ |A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D| \\
&- |A \cap B \cap C \cap D|.
\end{aligned}$$

As $A \cap B$ represents the set of the colorings having the first two rows red (and the 3 squares of the lowest row free to take any of the two colors), we have $|A \cap B| = 2^3$. Similarly, $|A \cap C| = 2^3$, $|B \cap D| = 2^3$ and $|C \cap D| = 2^3$. We have a different situation for $A \cap D$ and $B \cap C$: if two opposing 2-by-2 squares are red, this leaves uncertain only the colors of the two remaining 1-by-1 corners, therefore $|A \cap D| = |B \cap C| = 2^2$. The intersections of 3 of the sets leaves undecided only the color of one square, therefore each of these intersections has 2 elements. Finally, the intersection of all the four sets contains only the square completely colored in red, that is, one element. We obtain:

$$|A \cup B \cup C \cup D| = 4 \cdot 2^5 - (4 \cdot 2^3 + 2 \cdot 2^2) + 4 \cdot 2 - 1 = 95.$$

So from the 512 possible colorings, $512 - 95 = 417$ do not contain any 2-by-2 red square, so the probability of not having a 2-by-2 red square is $\frac{417}{512}$.

You can continue your solution for Problem 10 on this page.

Question 11: Conditional Probability [7 pts]

A man is known to tell the truth with probability $\frac{3}{4}$ (independent of the event outcome) and can only respond with "YES" or "NO". He rolls a fair six-sided die and is then asked whether the die landed on 6. He responds, "**YES**"! What is the probability that the die actually landed on 6? Justify your answer.

Answer: There are two scenarios in which the man responds YES: either he is being truthful, or he is not. Note that his truthfulness is independent of the die's outcome.

Let S be the event of rolling a six, $P(S) = \frac{1}{6}$.

Let $\bar{S}$ be the event of not rolling a six, $P(\bar{S}) = \frac{5}{6}$.

Let T be the event that the man tells the truth, regardless of the outcome, $P(T) = \frac{3}{4}$.

Let $\bar{T}$ be the event that the man does not tell the truth, regardless of the outcome, $P(\bar{T}) = \frac{1}{4}$.

Let M be the event that the man responds "YES" when asked if the die landed on 6. We want to find $P(S \mid M)$.

First Solution: Using the Bayes' theorem:

$$P(S \mid M) = \frac{P(M \mid S)P(S)}{P(M \mid S)P(S) + P(M \mid \bar{S})P(\bar{S})} = \frac{\frac{3}{4} \cdot \frac{1}{6}}{\frac{3}{4} \cdot \frac{1}{6} + \frac{1}{4} \cdot \frac{5}{6}} = \frac{3}{8}$$

Second Solution:

$$P(M) = P(S \cap T) + P(\bar{S} \cap \bar{T}) = \frac{1}{6} \cdot \frac{3}{4} + \frac{5}{6} \cdot \frac{1}{4} = \frac{8}{24}$$

$$P(S \cap M) = \frac{3}{4} \cdot \frac{1}{6} = \frac{3}{24}$$

$$P(S \mid M) = \frac{P(S \cap M)}{P(M)} = \frac{\frac{3}{24}}{\frac{8}{24}} = \frac{3}{8}$$

Problem 12: Graph Theory [12 pts]

(a) **[3 pts]** Consider a simple undirected graph with 6 nodes. Briefly explain (in at most 3 lines) why it is impossible for the nodes to have the following degree sequences:

(i) 5,4,3,2,2,0

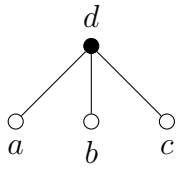
(ii) 4,4,3,2,1,1

(iii) 4,4,4,2,1,1

(b) **[4 pts]** How many distinct simple undirected graphs exist on 7 labeled vertices $\{a, b, c, d, e, f, g\}$ such that $\deg(a) \geq 4$?

(c) **[5 pts]** Consider a simple undirected graph G with n labeled vertices, where vertex i has degree d_i. Derive an expression for the total number of ordered paths of length 2 in G, in terms of the degrees $d_1, d_2, ..., d_n$. For this question, you may use the summation sign Σ to express your answer. Note that the origin and destination of the path must be different.

As an example, the ordered paths of length 2 in the graph below are: $\{(adb), (bda), (adc), (cda), (bdc), (cdb)\}$.



Answer: (a) (i) One node has a degree of 5 (means its connected to every other node) and one node has a degree of 0 (meanings not connected to anything) which is a contradiction.

(ii) The sum of the degrees is odd. So not possible.

(iii) Note that each of those nodes with degree 4, must be connected to at least one of the nodes with degree 1. (Why?)
So there are at least 3 edges going to those two nodes. So its not possible to have degree 1 for both.

(b) **First Solution**: There are 2^{21} possible graphs in total. Among these, exactly $\binom{6}{3} \times 2^{15}$ graphs satisfy $deg(a) = 3$; the remaining $2^{21} - \binom{6}{3} \times 2^{15}$ graphs are split equally between those with $deg(a) \geq 4$ and those with $deg(a) \leq 2$. This symmetry is due to each graph with $deg(a) \geq 4$ has a complement graph with $deg(a) \leq 2$. So, the answer to the question is:

$$\frac{2^{21} - \binom{6}{3} \times 2^{15}}{2} = 720896$$

Second Solution: We can count the number of graphs with $deg(a) = 4$, $deg(a) = 5$ and $deg(a) = 6$ directly:

$$\left(\binom{6}{4} + \binom{6}{5} + \binom{6}{6}\right) \cdot 2^{15} = 720896$$

(c) Note that each path consists of 3 nodes. By assuming each node i to be the middle node, the number of possible ordered paths is equal to the number of ways to choose an ordered pair of distinct neighbors of i. Since i has d_i neighbors, the number of ordered pairs of distinct neighbors is $d_i(d_i - 1)$. So the answer to the question is:

$$\sum_{i=1}^{n} d_i(d_i - 1)$$

Also accept:

$$2\sum_{i=1}^{n} \binom{d_i}{2}$$

You can continue your solution for Problem 12 on this page.

[scratch paper]

[scratch paper]

[scratch paper]